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NPS/001/018 – Technical Specification for Steel Poles

1. Purpose

The purpose of this document is to detail the technical specification for steel poles for use on the low voltage overhead line distribution networks of CE Electric UK.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NEDL/MS – 0/30	1.0	Technical specification for LV steel poles
NPS/001/018	1.0	Technical specification for LV steel poles

2. Scope

This document applies to steel poles for use on the low voltage overhead line networks of CE Electric UK.

3. Technical Requirements

3.1 General

Steel poles shall be manufactured from Grade S355JO steel in accordance with BS EN 10025 – 1: 2004 and be of tubular construction in accordance with BS EN 10210 – 2: 2006.

3.2 Welding

The sections shall be welded in accordance with BS EN 1011 – 1: 1998.

3.3 Galvanisation

Poles shall be galvanised in accordance with BS EN 1461: 1999 after fabrication. The coating shall be smooth, clean and of uniform thickness. The minimum thickness of the coating shall be 200 microns.

3.4 Dimensions

The dimensions are detailed on CE Electric UK drawing number 1.09.119.3501 sheet 2 as shown in Appendix A.

3.5 Mechanical Strength

Poles shall have a factor of safety of 2.5 based on the crippling loads of the steel. Section joints shall withstand a load of 2.5 x the safe working load applied 600mm from the top of the pole without exceeding the crippling load of the steel. These calculations are based on the appropriate sinking depth of the pole as detailed on drawing 1.09.119.3501 sheet 2.

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3.6 Pole Markings

Poles shall be marked with a 100mm horizontal line embossed at a distance 3m from the base of the pole. This will give an indication of how deep the pole has been sunk once it has been erected. Directly above this the pole shall be embossed with the year of manufacture. All fabrications and markings shall take place prior to galvanisation.

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4. References

4.1 External Documentation

Reference	Title
BS EN 10025 – 1: 2004	Hot rolled products of non-alloy structural steels
BS EN 10210 – 2: 2006	Hot finished structural hollow sections of non-alloy and fine grain structural steels. Tolerances, dimensions and sectional properties
BS EN 1011 – 1: 1998	Welding. Recommendations for welding of metallic materials
BS EN 1461: 1999	Hot dipped galvanised coatings on fabricated iron steel articles. Specifications and test methods

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
Steven Salkeld	Standards Engineer	Steven Salkeld	19/12/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Geoff Bolden	Purchasing Operations Manager	Geoff Bolden	20/12/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Ged Hammel	Overhead Standards Manager	Ged Hammel	20/12/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
Sean Johnson	DBD Administrator	Sean Johnson	11/01/08

6.5 Approval

Approval is given for the content of this document

		Sign	Date
Chris Holdsworth	Standards Manager	Chris Holdsworth	24/12/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
Carol Morgan	Head of Procurement	Carol Morgan	27/12/07

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Appendix 1 – Self Certification Conformance Declaration

Steel poles shall comply with the latest issues of the relevant international and British Standards.

This check sheet identifies the clauses in the aforementioned Standards relevant to steel poles for use on current United Kingdom designs of low voltage overhead lines.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Manufacturer:

Product Reference:

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'ENATS' as appropriate

Name:

Signature:

Date:

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BS EN 10210 – 2 – Tubular steel			
Clause / Sub-clause	Requirement	Conformance Code	Remarks
6.0	Tolerances		
8.0	Dimensions and sectional properties		

BS EN 1011 – 1 – Welding			
Clause / Sub-clause	Requirement	Conformance Code	Remarks
10.0	Fabrications		
11.0	Preparation of joints		
21.0	Traceability		
23.0	Inspection and testing		

BS EN 1461 – Galvanising			
Clause / Sub-clause	Requirement	Conformance Code	Remarks
4.0	General requirements		
5.0	Sampling		
6.0	Coating properties		
7.0	Certificate of compliance		

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Appendix 2 – Fabrication Drawing

